

Can a single X-ray observation tell a black hole from a neutron star in a source never seen in training?

Leakage-safe, pre-registered tests with RXTE, MAXI and NICER
(the XRB-pilot project, versions v1–v14)

Technical report

TODO: author name

Advisor: TODO: advisor name

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Chapter 1

Data sources and physical feature extraction

This chapter defines, once, every input quantity used anywhere in the project: where the raw data come from, how a raw file becomes a number, in which unit, by which function of the code, and what happens to non-positive or missing values. The phase chapters (Chapter 2 and ??????????) refer back here and only state differences. When the description in a repository report and the code differ, the code is followed and the difference is flagged with a `% CHECK` comment in the $\text{\LaTeX}$ source.

1.1 Instruments, data modes and catalogues

Table 1.1 lists every data product that enters a model or a test. All raw files were obtained by scripts; every download is recorded with its URL, size in bytes and SHA256 in `logs/downloads.jsonl` (50 773 records, ??). Raw data are not in git (`data/raw/` is ignored, large sets live outside the repository under `/xrb-pilot-data/`).

1.2 Source labels

The class label is a property of the *source*: $y = 1$ for a black hole (BH) and $y = 0$ for a neutron star (NS). Only labels with direct evidence are used for training and evaluation:

- **BH**: a dynamical mass function (radial-velocity curve of the companion). Versions v1–v4 used the dynamically confirmed systems of BlackCAT (?); v6 added high-mass X-ray binaries from ?; from v7 on the list of ? (25 systems in their Table 1) is used.
- **NS**: type-I X-ray bursts (thermonuclear burning on a solid surface) from ? (v1–v6) or the MINBAR catalogue (?) (v7c onwards), or coherent accretion-powered pulsations (?) (v6 external sources, v7c pulsar class; ? for HMXB pulsars in v7c).
- **BH candidates** (X-ray properties only, no dynamical mass) are never used as labels. In v3c they appear only as prediction targets and in a label-sensitivity analysis.

The label rules of every version are summarised in Table 1.2; the realised source lists are given in the phase chapters.

Table 1.1: Instruments, data modes and catalogues used in the project. “Versions” gives where the product enters a model or test. Band and resolution are those actually used by the code (not the instrument limits). Code: the function that reads or processes the product.

Product	Versions	What is used	Code
RXTE/PCA Standard-2 Stan- dard Products (xp*_s2.pha, _b2.pha, .rsp)	v1–v8, v10	129-channel total and <code>pcabackest</code> back- ground spectra summed over the PCUs listed in the <code>ROWID</code> keywords; 5–25 keV (3–25 keV in v3b); response for the same PCU combi- nation	<code>spectra.py: read_ pha, read_rsp, net_spectrum; 04_ fetch_process.py</code>
RXTE/PCA Standard-1 (pca/FS46_*)	v2–v10	0.125 s light curves per PCU (<code>XeCntPcu0–4</code>), plus array rates <code>VpCnt</code> , <code>RemainingCnt</code> , <code>VLECnt</code> ; 128 s rows of 1024 bins	<code>spectra.std1_ rates; v5lib.py; v6lib.py; v7lib.py</code>
RXTE/PCA event mode (pca/SE*.evt.gz)	v8c	Events with <code>TIME</code> , <code>PCUID</code> ; modes <code>E_{n}us_{m}M_0_{k}s</code> ; binned at 2^{-12} s	<code>v8lib.py: read_ events, hf_ features</code>
RXTE/HEXTE clus- ter B Standard Prod- ucts (xh*_s1.pha, _b1.pha)	v4b3	Net 25–40 and 40–60 keV count rates	<code>19_v4b3_hexte.py</code>
RXTE MissionLongData catalogues; HEASARC TAP xtemaster	v1–v8	Observation lists: <code>OBSID</code> , <code>TIME</code> , <code>EXPOSURE</code> , <code>STD1RATE</code> , <code>RA</code> , <code>DEC</code> , <code>NPCU</code>	<code>03_select_ observations.py, 20_v4b2_ select.py, 41_ v8b_select.py</code>
MAXI/GSC standard products, 1-day light curves	v7c, v8d, v9c	Photon fluxes in 2–4, 4–10, 10–20 keV	<code>38_v7c2_ select.py</code>
de Beurs et al. (2022) processed data (GitHub, commit 488b785)	v7c1	Their 44 sources, ten 20% subsamples, fea- tures (<code>RelInt</code> , <code>SC</code> , <code>HC</code>)	<code>37_v7c1_repro.py</code>
NICER/XTI cleaned events (ni*_Ompu7_cl.evt.gz); HEASARC TAP nicermastr	v9–v13	Events with <code>TIME</code> , <code>PI</code> , <code>DET_ID</code> ; 2–10 keV; 1/256 s bins, 128 s segments	<code>v9lib.py, v10lib.py, v11lib.py</code>
NICER unfiltered events (_ufa), NICER CALDB 3C50 library	v12, v13	Input of HEASoft <code>nibackgen3C50</code> (3C50 background model)	<code>v12_bkg3c50.sh, 58_v12_bkg.py, 62_v13_bkg.py</code>
MINBAR DR1 burst, observation and source tables	v7–v9	Burst start times and durations; burst-source positions	<code>v7lib.py: load_ minbar, gti_utc</code>
HI4PI (via HEASoft <code>nh</code>); XSPEC <code>tbabs</code>	v8d	Galactic N_H and band transmissions	<code>v8d_heasoft.sh, 45_v8d_nh.py</code>
CDS Sesame (SIM- BAD) positions	v4b2–v9	Source positions for label and pointing matches	<code>20_v4b2_ select.py, 28_ v6_select.py, 38_ v7c2_select.py</code>

Table 1.2: Source labelling and eligibility rules by version. “Pointing” is the maximum separation between an observation’s pointing and the source position used. Eligibility always also requires catalogue exposure ≥ 1000 s (RXTE: and $\text{STD1RATE} \geq 5 \text{ c s}^{-1} \text{ PCU}^{-1}$; NICER: `nicermastr` exposure ≥ 1000 s).

Version	BH rule	NS rule	Pointing / epoch / minimum
v1	4 BlackCAT dynamical BHs with ≥ 30 eligible pointings	4 Galloway+2008 bursters	0.1° from catalogue median; gain epoch 5, MJD 51677–54094
v2, v3	every BlackCAT Table 4 dynamical BH with a MissionLongData catalogue	every Galloway+2008 appendix-A source with a catalogue	0.1° ; epoch 5 (MJD 51677–55931); ≥ 10 eligible
v3c	16 BlackCAT candidates (prediction only)	–	as v2
v4b2	BlackCAT dynamical BHs matched by SIMBAD position	Galloway+2008 Table 3 bursters matched by position	all gain epochs; ≥ 10
v6 (external)	Casares & Jonker (2014) HMXB BHs	Patruno & Watts (2021) AMXPs (+ 4 slow pulsars as stress set)	0.1° from SIMBAD; all epochs; ≥ 10
v7b2	Cyg X-1, LMC X-1, LMC X-3 (Marcel+2026)	–	0.1° from SIMBAD; epoch 5; ≥ 10
v7c	Marcel+2026 Table 1	MINBAR bursters without pulsations (NPNS); pulsars (PSR) secondary	MAXI source within 0.1° (Sesame; 0.3° fallback); ≥ 100 daily points
v8b	unused Marcel+2026 BHs with RXTE data	–	0.1° from SIMBAD; all epochs; ≥ 3
v9–v13	Marcel+2026 Table 1 (Sesame positions)	MINBAR bursters, excluding entries with another entry within 0.05°	target within 0.05° ; ≥ 5

1.3 Observation eligibility and time-stratified sampling

For every source the eligible pointings are sorted by time and divided into K consecutive groups of (as nearly as possible) equal size, using `numpy.array_split` ($K = 15$ for RXTE, $K = 8$ for NICER). Inside each group the candidates are put in a random order generated by a seeded generator; the first candidate that passes the post-download quality rules (Section 1.4.5) is accepted. This keeps the sample spread over the source’s history (several outbursts, several states) and makes the choice independent of the spectrum. If a source has fewer than K eligible pointings, some groups are empty and all pointings are used. Algorithm 1 gives the procedure; the seeds are listed in Table 1.3.

Algorithm 1 Time-stratified candidate ranking and acceptance (`03_select_observations.py` + `04_fetch_process.py`; NICER: `47_v9_select.py`, `55_v11_select.py`, `60_v13_select.py`)

Input: eligible pointings $\{(o_i, t_i)\}$ of one source, number of groups K , seed σ , maximum tries M

- 1: sort pointings by time t_i ; split the index list into K consecutive groups $G_1, \dots, G_K$ (`array_split`)
- 2: `rng` $\leftarrow$ `default_rng`(σ)
- 3: **for** $b = 1, \dots, K$ **do**
- 4: $\pi \leftarrow \text{rng.permutation}(G_b)$ $\triangleright$ rank within the group
- 5: **for** $r = 1, \dots, \min(M, |G_b|)$ **do**
- 6: download and process o_{π_r} ;
- 7: **if** all quality rules pass **then** accept o_{π_r} ; **break**
- 8: **end if**
- 9: **end for**
- 10: **end for**

Table 1.3: Sampling parameters. “position” is the 1-based position of the source in the source list processed by the script. The global seed is `SEED = 42` (`config.py`).

Sample	groups K	seed σ	max. tries M
RXTE v1, v2, v3c, v7b2, v8b	15	<code>SEED + position</code>	6
RXTE v4b2 new sources	15	<code>SEED + 100 + i</code>	6
RXTE v4b1	one group per pointing (all eligible)	–	1
NICER v9	8	<code>SEED + position</code>	3
NICER v11	8	<code>SEED + 1000 + position</code>	3
NICER v13	8	<code>SEED + 2000 + position</code>	3

1.4 RXTE/PCA Standard-2 spectra

1.4.1 What the files contain

The Standard Products of every ObsID contain a total (source + background) spectrum `xp<obs>_s2.pha`, the `pcabackest` model background spectrum `xp<obs>_b2.pha` and the combined response `xp<obs>_rsp`. The header inspection of `05_two_source_check.py` (`reports/step2_fits_check.md`) established: `HDUCLAS2/3 = TOTAL/COUNT`, column `COUNTS` in counts, `POISSERR = F` with explicit `STAT_ERR`, `SYS_ERR = 0`, `BACKSCAL = AREASCAL = 1`, 129 channels, equal source and background exposures, and the PCUs summed listed in the `ROWIDn` keywords. The background statistical error is $\sqrt{\text{model counts}}$; the systematic uncertainty of the background

model is *not* included anywhere. Standard Products are *not* deadtime corrected. Figure 1.1 shows the two spectra used for this check.

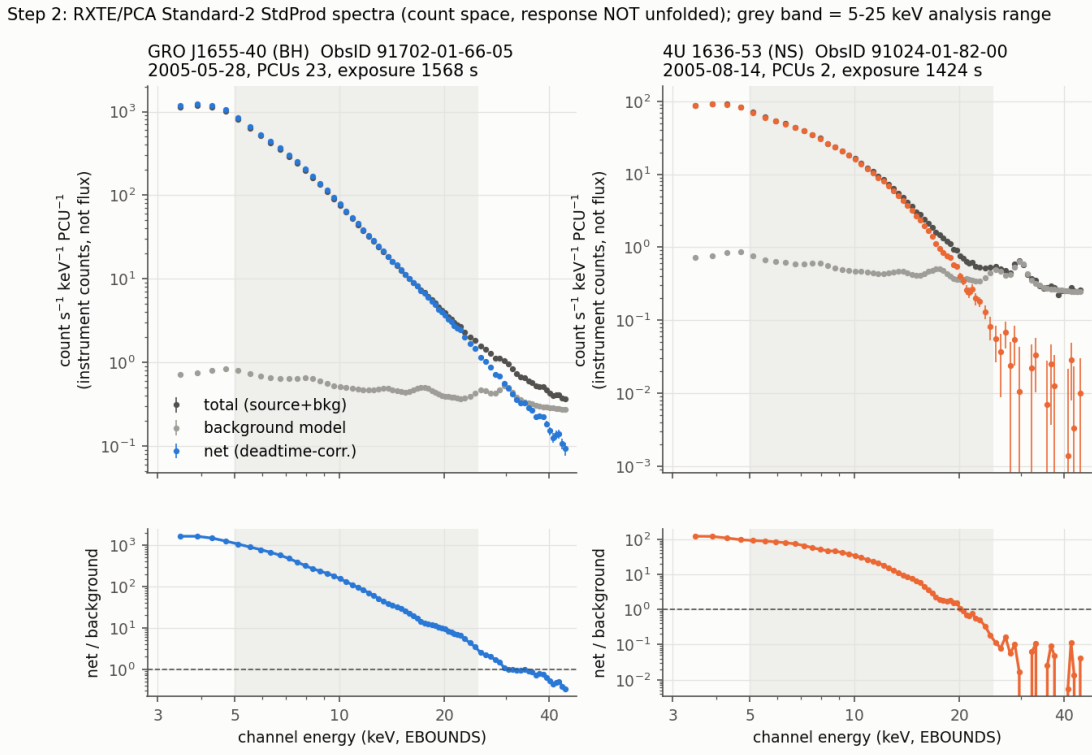


Figure 1.1: Data-flow validation on one BH (GRO J1655–40, ObsID 91702-01-66-05) and one NS (4U 1636–53, ObsID 91024-01-82-00). Top: total (dark grey), background model (light grey) and deadtime-corrected net (colour) count rates per keV per PCU against channel energy (EBOUNDS), log-log; the grey band is the 5–25 keV analysis range. Bottom: net/background ratio. The point is that the net spectrum is background-dominated only above ~ 30 keV for these bright sources, and that everything is in instrument count space (no response unfolding). Repo: `figures/step2_two_sources.png`, script `scripts/05_two_source_check.py`.

1.4.2 Background subtraction

For channel c of observation i , with source counts C_c^s and errors σ_c^s , background counts C_c^b and errors σ_c^b (function `spectra.net_spectrum`):

$$t_s = \text{EXPOSURE}_s \text{AREASCAL}_s, \quad t_b = \text{EXPOSURE}_b \text{AREASCAL}_b, \quad \kappa = \text{BACKSCAL}_s / \text{BACKSCAL}_b, \\ r_c^s = C_c^s / t_s, \quad e_c^s = \sigma_c^s / t_s, \quad r_c^b = \kappa C_c^b / t_b, \quad e_c^b = \kappa \sigma_c^b / t_b, \quad (1.1)$$

$$r_c^{\text{net}} = D (r_c^s - r_c^b), \quad e_c^{\text{net}} = D \sqrt{(e_c^s)^2 + (e_c^b)^2}, \quad (1.2)$$

in count s⁻¹ (summed over the PCUs of the spectrum), where $D = 1/(1 - \text{DTF})$ is the deadtime correction factor (Section 1.4.3). This is the OGIP/XSPEC convention: the same quantity that XSPEC uses after `data`, `backgrnd` and `response`. Non-positive net rates (possible where the source is faint, mainly at $E > 20$ keV in soft spectra) are kept unchanged: no absolute value, no clipping, no deletion.

1.4.3 Deadtime

v1 (lower bound). Only the good-xenon term was available from Standard-2:

$$\text{DTF}^{(\text{v1})} = 10^{-5} \text{ s} \times \frac{\sum_c C_c^s}{t_s N_{\text{PCU}}}, \quad (1.3)$$

i.e. the total (source + background) count rate per PCU times $10 \mu\text{s}$. Median 0.31% and maximum 6% in the v1 sample (`data/observations.csv`, column `dtf_xe`).

v2 onwards (RXTE GOF recipe from Standard-1). The Standard-1 files of the ObsID are read and the mean rates inside the spectrum's good time intervals (GTIs) are computed from all 0.125 s bins whose centre lies in a GTI (`spectra.std1_rates`): per-PCU good-xenon rates X_{e_p} ($p = 0, \dots, 4$) and array rates V_p (propane), Rem (coincident events) and VLE (very large events). With N_{on} the number of PCUs with $X_{e_p} > 1 \text{ c s}^{-1}$ (`spectra.std1_dtf`):

$$\text{DTF}_p = 10^{-5} \left(X_{e_p} + \frac{V_p + \text{Rem}}{N_{\text{on}}} \right) + 6 \times 10^{-5} \frac{\text{VLE}}{N_{\text{on}}}, \quad \text{DTF} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \text{DTF}_p, \quad D = \frac{1}{1 - \text{DTF}}, \quad (1.4)$$

with rates in count s^{-1} and $\mathcal{P}$ the PCUs of the spectrum. A version with $1.5 \times 10^{-4} \text{ s}$ per VLE event (quoted elsewhere on the same GOF page) was stored only as a sensitivity number (`dtf_std1_vle150us`). In the v2 sample the Standard-1 coverage of the spectrum GTIs has median 99% (minimum 88%), the DTF has median 1.38% and maximum 8.8%; no observation exceeded the 10% rule. Because D multiplies the whole spectrum, it changes intensities (representation A) by 1–9% and leaves shapes (B, H) unchanged.

1.4.4 Common energy grid and rebinning

Gain drifts change the channel-to-energy relation slightly between observations, so every spectrum is rebinned onto one fixed grid defined by a reference observation (`REFERENCE_OBS = 91702-01-66-05`). Let $\{E_k^{\text{lo}}, E_k^{\text{hi}}\}$ be the EBOUNDS of the reference response. The grid edges are the union of all EBOUNDS edges between the edge closest to 5 keV and the edge closest to 25 keV: $e_0 = 4.908, \dots, e_{45} = 25.008 \text{ keV}$ (45 bins, widths 0.41–0.86 keV; Table 1.4). For v3b the lower limit is the edge closest to 3 keV, giving 50 bins from 2.883 keV; the 45 bins above 4.908 keV are identical. For an observation with its own EBOUNDS, the overlap matrix (`spectra.overlap_matrix`)

$$W_{jk} = \frac{\max(0, \min(E_k^{\text{hi}}, e_{j+1}) - \max(E_k^{\text{lo}}, e_j))}{E_k^{\text{hi}} - E_k^{\text{lo}}} \quad (1.5)$$

is the fraction of channel k that lies in grid bin j (counts assumed uniform within a channel). The rebinned net rate per keV per PCU and its error are

$$x_{ij} = \frac{\sum_k W_{jk} r_{ik}^{\text{net}}}{N_{\text{PCU},i} \Delta e_j}, \quad \sigma_{ij} = \frac{\sqrt{\sum_k W_{jk}^2 (e_{ik}^{\text{net}})^2}}{N_{\text{PCU},i} \Delta e_j}, \quad \Delta e_j = e_{j+1} - e_j, \quad (1.6)$$

in $\text{count s}^{-1} \text{ keV}^{-1} \text{ PCU}^{-1}$ (array `rate` and `err` of `features.npz`). Division by the number of PCUs is justified because the effective area grows linearly with the number of PCUs ($\approx 1070 \text{ cm}^2$ per PCU at 6 keV). The error formula treats channels as independent; two grid bins that share one channel are therefore weakly correlated, which is ignored. The total (source + background) and background spectra are rebinned in the same way (`srates`, `brate`). The gain drift within epoch 5 is at most 0.85 channel (median 0.17) relative to the reference.

Table 1.4: The 45-bin common energy grid (keV), from the EBOUNDS of observation 91702-01-66-05 (`data/v2/processed/features.npz`, key `edges`; identical in every RXTE sample). Bin j spans $[e_j, e_{j+1})$. The v3b grid adds five bins below: 2.883, 3.287, 3.691, 4.096, 4.502, 4.908 keV.

j	e_j	j	e_j	j	e_j	j	e_j	j	e_j
0	4.908	10	9.001	20	13.149	30	17.346	40	21.588
1	5.315	11	9.414	21	13.566	31	17.768	41	22.014
2	5.722	12	9.827	22	13.985	32	18.191	42	22.441
3	6.130	13	10.240	23	14.403	33	18.614	43	23.295
4	6.538	14	10.654	24	14.822	34	19.038	44	24.151
5	6.948	15	11.069	25	15.242	35	19.462	45	25.008
6	7.357	16	11.484	26	15.662	36	19.886		
7	7.767	17	11.899	27	16.082	37	20.311		
8	8.178	18	12.315	28	16.503	38	20.736		
9	8.589	19	12.732	29	16.924	39	21.162		

1.4.5 Summary quantities, errors and quality rules

For each observation (`04_fetch_process.py`):

$$F_i = \sum_{j=0}^{44} x_{ij} \Delta e_j \quad [\text{count s}^{-1} \text{PCU}^{-1}], \quad \sigma_{F,i} = \left(\sum_j (\sigma_{ij} \Delta e_j)^2 \right)^{1/2}, \quad S/N_i = F_i / \sigma_{F,i}, \quad (1.7)$$

$$f_i^{\text{bkg}} = \frac{\sum_j x_{ij}^b \Delta e_j}{\sum_j x_{ij}^s \Delta e_j} \quad (\text{background fraction, 5–25 keV}).$$

An observation is accepted only if all of the following hold (otherwise the next candidate of the same time group is tried): all files present and readable, 129 channels, EBOUNDS cover 5–25 keV; spectrum exposure ≥ 1000 s; $S/N \geq 10$; $DTF \leq 0.10$; header position `RA_OBJ/DEC_OBJ` within 0.1° of the source position; all values finite. $F_i > 0$ is guaranteed by the S/N rule, so the shape normalisation (Eq. (1.10)) is always defined.

1.4.6 Response check and response-normalised spectra

A fixed photon spectrum $N(E) = E^{-2}$ (photon index $\Gamma = 2$, no absorption, norm $1 \text{ ph cm}^{-2} \text{ s}^{-1} \text{ keV}^{-1}$ at 1 keV) is folded through each observation’s response matrix $R(E, c)$ (`spectra.powerlaw_folded`) and rebinned like the data:

$$P_{ic} = \sum_{\ell} R_i(E_{\ell}, c) \int_{E_{\ell}^{\text{lo}}}^{E_{\ell}^{\text{hi}}} E^{-2} dE, \quad p_{ij} = \frac{\sum_c W_{jc} P_{ic}}{N_{\text{PCU},i} \Delta e_j} \quad (\text{key p12}). \quad (1.8)$$

The across-observation scatter of $p_{ij} / \text{median}_i p_{ij}$ measures how much the instrument response alone changes the count spectrum: median 3.7%, maximum 11–12.7%, driven by the PCU combination (Figure 1.2). BH and NS have similar PCU-combination mixtures (mean tilt 2.0% vs 2.4% in v1), so this is treated as noise rather than as a class shortcut. From v4b2 on, p_{ij} also defines the response-normalised representations R and H_R (Section 1.5), which can be compared across gain epochs.

1.4.7 Independent check with XSPEC

For 12 randomly chosen v2 observations (seed 42; 10 sources; net rates $5\text{--}950 \text{ cs}^{-1}$), HEASoft 6.37.1/XSPEC (`data, backgrnd, response, ignore **-5.0 25.0-**, tclout rate`) and

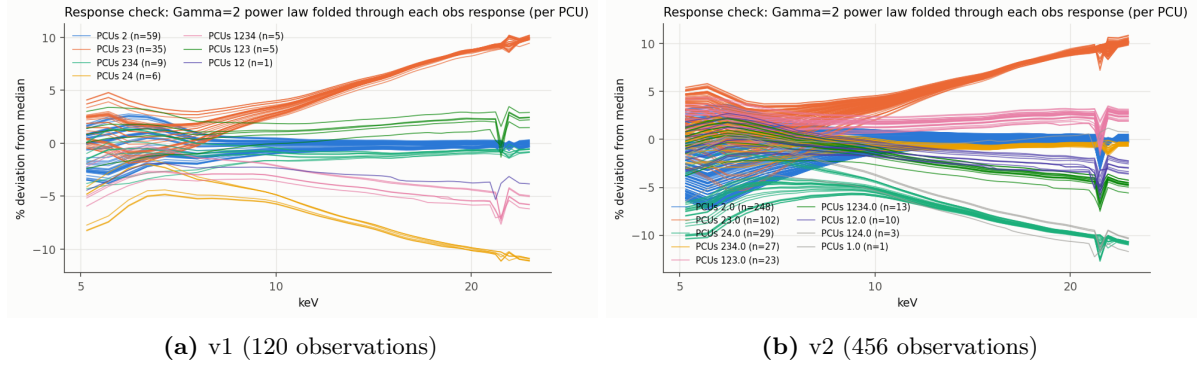


Figure 1.2: Response check: a fixed $\Gamma = 2$ power law folded through each observation’s response, per PCU, shown as percentage deviation from the median over observations, against energy; colour = PCU combination. The spread (a few per cent, up to $\sim 10\%$ at 25 keV) is set by which PCUs were on, not by the source class. Repo: `figures/step3_response_variation.png`, `figures/v2/step3_response_variation.png`; script `scripts/06_dataset_checks.py`.

Eq. (1.2) with $D = 1$ summed over the same noticed channels (10–52 or 10–53) agree to 3×10^{-10} (rate) and 4×10^{-10} (error) relative (`results/v2/xspec_crosscheck.csv`, `scripts/10_xspec_crosscheck.py`). This validates Eqs. (1.1) and (1.2); it does not validate the deadtime, the rebinning, or the background model itself.

1.5 Spectral representations

All representations are computed per observation from the rebinned spectrum x_i ; none uses information from other observations (the training-set standardisation of ?? is a separate step inside the model). Code: `07_train_eval.py`, `v3lib.v2_representations`, `12_v3b_extended_band.py`, `reps_R` in `22_v4b2_analysis.py`.

A (intensity kept). With scale $a = 0.1 \text{ count s}^{-1} \text{ keV}^{-1} \text{ PCU}^{-1}$ (`ASINH_SCALE`),

$$A_{ij} = \text{asinh}(x_{ij}/a), \quad j = 0, \dots, 44. \quad (1.9)$$

$\text{asinh}(u) \approx \text{sgn}(u) \ln(2|u|)$ for $|u| \gg 1$ and $\approx u$ near 0, so A is log-like for bright bins and defined for the (rare) negative bins. A keeps the absolute count rate, which depends on distance, absorption and accretion rate as well as on the compact object.

B (shape).

$$B_{ij} = x_{ij}/F_i \quad [\text{keV}^{-1}], \quad \sum_j B_{ij} \Delta e_j = 1. \quad (1.10)$$

H (two count-space colours). Four bands are formed from the grid bins whose geometric centre $\bar{e}_j = \sqrt{e_j e_{j+1}}$ lies in a nominal interval (`v3lib.band`); the exact bin ranges are in Table 1.5:

$$C_{i,m} = \sum_{j \in b_m} x_{ij} \Delta e_j \quad [\text{count s}^{-1} \text{ PCU}^{-1}], \quad \mathbf{h}_i = (h_{i1}, h_{i2}) = \left(\frac{C_{i,2}}{C_{i,1}}, \frac{C_{i,4}}{C_{i,3}} \right). \quad (1.11)$$

h_1 (“soft colour”, ≈ 7 –10 over 5–7 keV) and h_2 (“hard colour”, ≈ 16 –25 over 10–16 keV) are linear ratios. The pre-registration originally specified $\log_{10}$ colours; before any v2 model was trained, two BH spectra (95409-01-29-00 GX 339–4, 60113-01-34-00 XTE J1650–500) were

found with a slightly negative 16–25 keV net sum, so the linear ratio was adopted (logged in `reports/decision_log.md`). Denominators C_1, C_3 are positive for every observation (asserted in the code); numerators may be ≤ 0 and are kept.

HI, H+B, HI+A. $\text{HI}_i = (h_{i1}, h_{i2}, \log_{10} F_i)$; $\text{H+B}_i = (\mathbf{h}_i, B_{i0}, \dots, B_{i44})$ (47 features); $\text{HI+A}_i = (\text{HI}_i, A_{i0}, \dots, A_{i44})$ (48 features).

v3b (3–25 keV). On the 50-bin grid: $F_i^{(3)} = \sum_j x_{ij} \Delta e_j$ over all 50 bins; $\text{B3}_{ij} = x_{ij}/F_i^{(3)}$; $\text{H3}_i = (C_{i,0}/C_{i,1}, C_{i,2}/C_{i,1}, C_{i,4}/C_{i,3})$ where $C_{i,0}$ sums the five bins with centre in $[2.8, 5)$ keV (2.883–4.908 keV). The soft colour has 3–5 keV in the numerator and a denominator that is always positive.

R and H_R (response-normalised; v4b2 onwards). With p_{ij} of Eq. (1.8):

$$R_{ij} = \frac{x_{ij}}{p_{ij}}, \quad P_{i,m} = \sum_{j \in b_m} p_{ij} \Delta e_j, \quad \mathbf{h}_i^R = \left(\frac{C_{i,2}/P_{i,2}}{C_{i,1}/P_{i,1}}, \frac{C_{i,4}/P_{i,4}}{C_{i,3}/P_{i,3}} \right). \quad (1.12)$$

H_R is the colour of the data relative to the colour a $\Gamma = 2$ power law would have through the same response; it removes first-order response differences (e.g. between gain epochs). Its predictions agree with H at Spearman 0.994 (??).

Table 1.5: The four colour bands of H (and H_R): grid bins whose geometric centre lies in the nominal interval, and the actual energy range they cover (computed from `edges`).

Band	nominal interval (centre rule)	bins j	n bins	actual range [keV]
b_1	[5, 7)	0–4	5	4.908–6.948
b_2	[7, 10)	5–11	7	6.948–9.827
b_3	[10, 16)	12–26	15	9.827–16.082
b_4	[16, 25.1)	27–44	18	16.082–25.008
b_0 (v3b)	[2.8, 5)	5 bins of the 50-bin grid	5	2.883–4.908

1.6 HEXTE high-energy colours (v4b3)

Only HEXTE cluster B is used (`19_v4b3_hexte.py`), because it rocked between source and background until 2009-12-14 16:10 UT (MJD 55179.6736); observations on or after that date, or without cluster-B files, have no HEXTE features (60 of the 456 v2 observations). Header keyword `DEADAPP = T` (deadtime already applied) is required. The net cluster-B spectrum follows Eqs. (1.1) and (1.2) with $D = 1$; band rates $F_X(25\text{--}40)$ and $F_X(40\text{--}60)$ (count s^{−1} per cluster) use the overlap matrix of Eq. (1.5) on the RMF EBOUNDS. With $F_{\text{PCA}}(16\text{--}25) = C_{i,4}$ of Eq. (1.11):

$$X_1 = \frac{F_X(25\text{--}40)}{F_{\text{PCA}}(16\text{--}25)}, \quad X_2 = \frac{F_X(40\text{--}60)}{F_X(25\text{--}40)}. \quad (1.13)$$

X_1 is a cross-instrument count ratio and is only used as a feature. A colour is set to missing when its denominator has $\text{S/N} < 3$ (X_1 : 3 observations; X_2 : 144, of which 23 BH). Missing colours are filled with the training-fold median, and one 0/1 indicator per colour is appended (inside the LOSO loop; `hx_features`); H+X therefore has 6 columns.

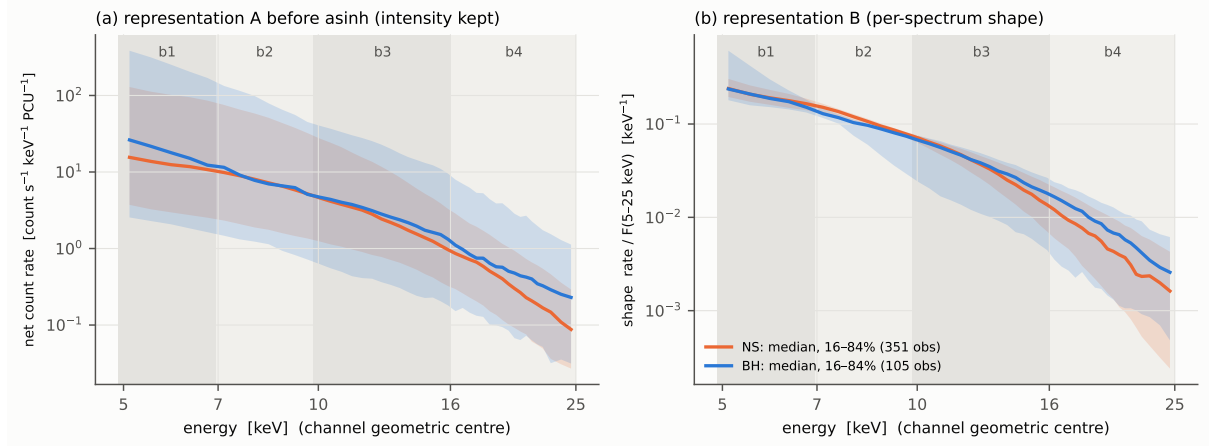


Figure 1.3: Median (line) and 16–84% range (shading) of the v2 count-space spectra, BH (blue, 105 observations) and NS (orange, 351 observations). (a) net rate per keV per PCU (the quantity transformed by asinh in representation A); (b) per-spectrum shape B. Grey and light bands mark the four colour bands b_1 – b_4 of Table 1.5. Non-positive bins are omitted from the percentiles. The class medians differ mainly in the overall slope above 10 keV and overlap strongly, which is why two colours capture most of the shape information (Chapter 2). Generated for this report from `data/v2/processed/features.npz` (`report_en/figures/generated/make_figures.py`).

1.7 Type-I X-ray bursts

Type-I bursts are seconds-long thermonuclear flashes that only NSs produce; an observation containing one carries a label shortcut and a distorted spectrum. Three procedures are used.

Standard-1 burst detector (v4; `17_v4_burst_detect.py`). The 1 s light curve $r(t)$ is the sum of good-xenon counts of the spectrum’s PCUs over complete 1 s bins inside the GTIs. With μ, s the mean and standard deviation of all bins: candidates are bins with $r > \mu + 4s$; candidates closer than 30 s are merged into one event. For an event with peak r_{pk} at t_{pk} and persistent level r_0 (median of the 60 s before the event, or of the whole curve if fewer than 60 bins), amplitude $A = r_{\text{pk}} - r_0$ and levels $L_q = r_0 + qA$:

$$\text{burst} \iff r_{\text{pk}} \geq 1.5 r_0 \wedge t_{\text{rise}} \leq 10 \text{ s} \wedge 3 \leq t_{1/2} \leq 300 \text{ s} \wedge n_{25} \geq 3, \quad (1.14)$$

where $t_{\text{rise}} = t_{\text{pk}} - \max\{t < t_{\text{pk}} : r(t) < L_{0.25}\}$, $t_{1/2} = \min\{t > t_{\text{pk}} : r(t) < L_{0.5}\} - t_{\text{pk}}$, and n_{25} is the number of contiguous 1 s bins around the peak above $L_{0.25}$. The burst interval runs from the last time before the peak below $L_{0.25}$ to the first time after it below $L_{0.1}$. Thresholds are `V4_BURST` in `config.py`, following the search criterion of ?. Every candidate was also inspected by eye (`results/v4/burst_visual.csv`).

MINBAR matching (v7b1; `v7lib.load_minbar, gti_utc, minbar_flag`). The spectrum GTIs (STDGTI, mission elapsed time in TT) are converted to $\text{MJD}(\text{TT}) = \text{MJDREF} + t/86400$ and then to UTC with `astropy` ($\text{TT} - \text{UTC} = 64.18 \text{ s}$; round-trip error $< 1 \text{ ms}$). A MINBAR burst occupies $[T, T + \text{dur}]$ (dur missing $\Rightarrow 300 \text{ s}$). Primary rule: an observation is contaminated if a burst interval overlaps a GTI *and* the burst has the same ObsID or originates within 1° (PCA field of view) of the source. The literal pre-registered rule (any burst from any origin) is kept as a sensitivity set.

NICER burst rule (v9–v13; `v9lib.burst_events`). The same Eq. (1.14) is applied to the 2–10 keV 1 s light curve built from events inside the good intervals (Section 1.10); every 128 s

segment that overlaps $[t_{\text{start}} - 20 \text{ s}, t_{\text{end}} + 200 \text{ s}]$ of an automatically detected burst is removed.

1.8 Low-frequency timing from RXTE Standard-1 (v5–v10)

Figure 1.4 summarises the chain from light curve to features; this section gives the formulae.

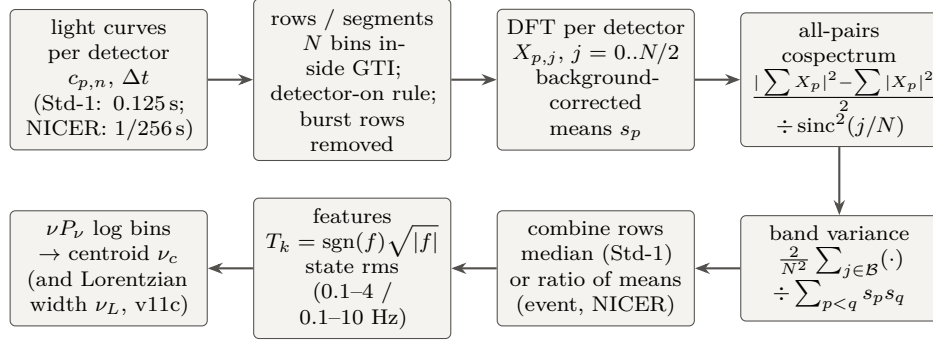


Figure 1.4: Timing-feature pipeline shared by RXTE Standard-1 (v5–v10), RXTE event mode (v8c) and NICER (v9–v13). Differences between the instruments are in the bin width, the segment length, the “detector” (PCU or NICER MPU), whether rows are combined by the median or by the ratio of means, and the frequency bands (Table 1.6). Drawn for this report.

1.8.1 Rows, detectors and background

A Standard-1 file stores, for each 128 s row, $N = 1024$ counts per PCU in $\Delta t = 0.125 \text{ s}$ bins. Only rows lying completely inside a spectrum GTI are used (start $\geq$ GTI start -1 ms and end $\leq$ GTI stop $+1 \text{ ms}$; `v5lib.std1_rows`). PCU p is *on* in a row if each of the eight 128-bin blocks (16 s) has at least one count (`v5lib.pcus_on`). The background count rate per PCU is taken from the Standard-2 background spectrum over all channels, $b = \sum_c C_c^b / (t_b N_{\text{PCU},b})$ counts s^{-1} (assumed equal for all PCUs). For a detector group or a single PCU with counts c_n and n_{det} detectors, the mean *source* counts per bin is $s = \bar{c} - n_{\text{det}} b \Delta t$; a row is invalid if any $s \leq 0$.

1.8.2 Fourier conventions and the binning correction

The discrete Fourier transform is $X_j = \sum_{n=0}^{N-1} c_n e^{-2\pi i j n / N}$, $j = 0, \dots, N/2$ (`numpy.fft.rfft`), frequency $\nu_j = j / (N \Delta t)$ (for 128 s rows, $\nu_j = j / 128 \text{ Hz}$). Parseval’s relation gives $\frac{1}{N} \sum_n (c_n - \bar{c})^2 = \frac{1}{N^2} \sum_{j \neq 0} |X_j|^2 \approx \frac{2}{N^2} \sum_{j=1}^{N/2} |X_j|^2$, so $\frac{2}{N^2} \sum_{j \in \mathcal{B}} |X_j|^2$ is the variance (in counts² per bin²) contributed by band $\mathcal{B}$. Integrating counts over bins of width Δt multiplies the Fourier amplitude by $\text{sinc}(j/N) = \sin(\pi j/N) / (\pi j/N)$ (? , Eq. 2.19); every power is divided by $\text{sinc}^2(j/N)$.

1.8.3 Two-group cospectrum (v5)

The PCUs that are on are sorted; those in even positions form group A, those in odd positions group B (at least two PCUs are required). With x_n, y_n the summed counts of the groups, X_j, Y_j their transforms, and s_x, s_y their background-corrected mean source counts per bin (`v5lib.row_features`):

$$V_{\mathcal{B}} = \frac{2}{N^2} \sum_{j \in \mathcal{B}} \frac{\text{Re}(X_j Y_j^*)}{\text{sinc}^2(j/N)}, \quad f_{\mathcal{B}} = \frac{V_{\mathcal{B}}}{s_x s_y}. \quad (1.15)$$

Because the Poisson (and deadtime) noise of different detectors is uncorrelated, the expected value of the cospectrum $\text{Re}(X_j Y_j^*)$ in the absence of source variability is zero (??); no Poisson level has to be modelled. $f_{\mathcal{B}}$ is the fractional variance (rms² normalised to the source mean) in band $\mathcal{B}$. Per observation, the *median* over valid rows is taken (robust to rare rows with flares or bursts); fewer than 3 valid rows $\Rightarrow$ missing. Features are signed square roots,

$$T_k = \text{sgn}(\text{median}_r f_{k,r}) \sqrt{|\text{median}_r f_{k,r}|}, \quad k \in \{1, 2, 3\}, \quad T_{\text{tot}} = \text{sgn}(F) \sqrt{|F|}, \quad F = \sum_k \text{median}_r f_{k,r}, \quad (1.16)$$

(fractional rms, which may be slightly negative because of noise). The bands are listed in Table 1.6. As an implementation check the conventional auto-power estimator $(|Z_j|^2 - N\bar{z})/\text{sinc}^2(j/N)$ of $z = x + y$ (Poisson level $N\bar{z}$, no deadtime) was also computed; for 276 S1 observations with 20–500 source $\text{cs}^{-1} \text{PCU}^{-1}$ the two versions of T_2 agree at Spearman 0.970.

1.8.4 All-pairs estimator and single-PCU hybrid (v6)

Using every pair of the n PCUs that are on, rather than one A/B split, reduces the noise (`v6lib.allpairs_row`). With per-PCU transforms $X_{p,j}$ and source means s_p :

$$\sum_{p < q} \text{Re}(X_{p,j} X_{q,j}^*) = \frac{|\sum_p X_{p,j}|^2 - \sum_p |X_{p,j}|^2}{2}, \quad f_{\mathcal{B}} = \frac{\frac{2}{N^2} \sum_{j \in \mathcal{B}} \frac{|\sum_p X_{p,j}|^2 - \sum_p |X_{p,j}|^2}{2 \text{sinc}^2(j/N)}}{\frac{(\sum_p s_p)^2 - \sum_p s_p^2}{2}}. \quad (1.17)$$

Rows with a single PCU on use the auto-power estimator $f_{\mathcal{B}} = \frac{2}{N^2} \sum_{j \in \mathcal{B}} \frac{|X_j|^2 - N\bar{c}}{\text{sinc}^2(j/N)} / s^2$, but only if the total count rate is $\leq 500 \text{cs}^{-1}$ (where the deadtime-free Poisson level is adequate); otherwise the row is invalid. T_k^* is the signed square root of the median over all valid rows (pair rows and single rows together). In synthetic tests the noise variance drops by 31%, 33% and 41% for 3, 4 and 5 PCUs (theory 33%, 33%, 40%) without bias; missing values over 8350 observations fall from 942 to 341.

1.8.5 Characteristic frequency ν_c (v6)

Nine logarithmic Fourier-index bins are formed from $\text{unique}(\text{round}(\text{geomspace}(1, 512, 10))) = \{1, 2, 4, 8, \dots, 512\}$ as $[1, 1], [2, 3], [4, 7], \dots, [256, 511]$ (Table 1.7). For each bin b , P_b is the median over rows of the fractional variance (Eq. (1.17)) in that bin; the bin centre is $\nu_b = \sqrt{j_{\text{lo}} j_{\text{hi}}}/128 \text{Hz}$ and its width in $\ln \nu$ is $\Delta_b = \ln((j_{\text{hi}} + 0.5)/(j_{\text{lo}} - 0.5))$. The νP_ν centroid is (`v6lib.nu_centroid`)

$$\ln \nu_c = \frac{\sum_b w_b \ln \nu_b}{\sum_b w_b}, \quad w_b = \frac{\max(P_b, 0)}{\Delta_b}, \quad (1.18)$$

missing if $T_{\text{tot}}^* < 0.05$ or $\sum_b w_b \leq 0$. Since P_b/Δ_b is the power per unit $\ln \nu$ (i.e. νP_ν), ν_c is the frequency around which most of the variability power per logarithmic frequency interval lies: a model-free proxy for a break or characteristic frequency. Synthetic sinusoids at 0.05, 0.5 and 3 Hz are recovered within one log bin.

1.8.6 State index (v7a)

The variability state of an observation is classified by its fractional rms in 0.1–4 Hz ($j = 13\text{--}511$), computed with Eq. (1.17) from rows with ≥ 2 PCUs only (no single-PCU rows, because their Poisson level would need a deadtime model), after removing every row that overlaps a burst

Table 1.6: Fourier-index bands and their frequencies. Standard-1 and NICER use 128 s rows or segments ($\nu_j = j/128$ Hz); RXTE event mode uses 16 s segments ($\nu_j = j/16$ Hz). From `config.py`.

Band	index j	frequency [Hz]	data	config name
T_1	2–12	0.0156–0.094	Std-1, NICER	V5_BANDS_J, V9_T_BANDS_J
T_2	13–128	0.102–1.000	Std-1, NICER	
T_3	129–511	1.008–3.992	Std-1, NICER	
state (v7a)	13–511	0.102–3.992	Std-1	V7_RMS_J
state (v9+)	13–1280	0.102–10.0	NICER	V9_STATE_J
white monitor	12288–16383	96–128	NICER	V9_WHITE_J
LC	16–63	1.0–3.94	event mode	V8_HF_BANDS_J
HF1	64–1023	4.0–63.9	event mode	
HF2	1024–8191	64–512	event mode	
HF3	8192–16383	512–1024	event mode	
NULL	24576–32767	1536–2048	event mode	

Table 1.7: Bins of the νP_ν centroid. Left: RXTE Standard-1 (v6; 9 bins, ν_b and Δ_b of Eq. (1.18)). Right: NICER (v9–v13; 13 octave bins $j \in [2^k, 2^{k+1} - 1]$, centre $2^{k+0.5}/128$ Hz). Computed from the code (`v6lib.py`, `v9lib.py`).

Standard-1 (v6)				NICER (v9+)		
$[j_{lo}, j_{hi}]$	ν_b [Hz]	range [Hz]	Δ_b	k	$[j_{lo}, j_{hi}]$	ν_k [Hz]
[1, 1]	0.0078	0.0039–0.0117	1.099	0	[1, 1]	0.011
[2, 3]	0.0191	0.0117–0.0273	0.847	1	[2, 3]	0.022
[4, 7]	0.0413	0.0273–0.0586	0.762	2	[4, 7]	0.044
[8, 15]	0.0856	0.0586–0.121	0.726	3	[8, 15]	0.088
[16, 31]	0.174	0.121–0.246	0.709	4	[16, 31]	0.177
[32, 63]	0.351	0.246–0.496	0.701	5	[32, 63]	0.354
[64, 127]	0.704	0.496–0.996	0.697	6	[64, 127]	0.707
[128, 255]	1.411	0.996–1.996	0.695	7	[128, 255]	1.414
[256, 511]	2.826	1.996–3.996	0.694	8	[256, 511]	2.828
				9–12	[512, 1023]–[4096, 8191]	5.66, 11.3, 22.6, 45.3

interval $[t_{\text{start}} - 20 \text{ s}, \text{end}]$ (MINBAR field-of-view rule and the v4 detector; `v7lib.state_rms`). With $\rho = \text{sgn}(\text{median}_r f) \sqrt{|\text{median}_r f|}$:

$$\text{state} = \begin{cases} \text{hard-like} & \rho > 0.10 \\ \text{soft-like} & \rho < 0.075 \\ \text{intermediate} & \text{otherwise} \end{cases} \quad (\text{unknown if fewer than 3 rows}), \quad (1.19)$$

the thresholds of ρ , Table 2, which are defined for 0.1–10 Hz; the 4 Hz Nyquist limit of Standard-1 makes ρ slightly low and the hard-like class conservative. NICER (v9 onwards) uses the full 0.1–10 Hz band. “Soft-like/hard-like” are operational variability classes, not full spectral-state assignments.

1.9 High-frequency timing from RXTE event mode (v8c)

An event file qualifies if DATAMODE matches `^E_\d+us_\d+M_0_\d+s$`, TDES2 contains the six anode chains `X1L^X1R^X2L^X2R^X3L^X3R`, `TIMEDEL` $\leq 2^{-12} \text{ s}$ and a PCUID column exists (`v8lib.mode_qualifies`); if several modes qualify, the one with the largest total size is used. Events are binned at $\Delta t = 2^{-12} \text{ s}$ (Nyquist 2048 Hz) in 16 s segments ($N = 65536$) lying completely in (spectrum GTI) $\cap$ (event-file GTI) and not overlapping a burst interval. A PCU is on if each of the 16 one-second sub-intervals has an event; ≥ 2 PCUs are required. For segment r , with Eq. (1.17), $\text{num}_{\mathcal{B},r} = \frac{2}{N^2} \sum_{j \in \mathcal{B}} \text{cross}_j$ and $\text{den}_r = \sum_{p < q} s_p s_q$. The pre-registered per-observation estimator was the mean of the per-segment ratios; because single 16 s segments of faint sources give unstable ratios (one Aql X-1 segment gave -195), it was replaced, after the implementation check IC4 and before any model, by the ratio of means with a delta-method standard error (`v8lib.hf_summary`):

$$\hat{R}_{\mathcal{B}} = \frac{\sum_r \text{num}_{\mathcal{B},r}}{\sum_r \text{den}_r}, \quad \widehat{\text{SE}}(\hat{R}_{\mathcal{B}}) = \frac{1}{\sum_r \text{den}_r} \sqrt{\frac{n}{n-1} \sum_r (\text{num}_{\mathcal{B},r} - \hat{R}_{\mathcal{B}} \text{den}_r)^2}, \quad (1.20)$$

with n segments; features are $\text{sgn}(\hat{R}) \sqrt{|\hat{R}|}$ for LC, HF1, HF2, HF3. Quality: $n \geq 10$ and $|z_{\text{NULL}}| = |\hat{R}_{\text{NULL}} / \widehat{\text{SE}}| \leq 5$ in the empty 1536–2048 Hz band. A post hoc correction subtracts the null-band power scaled by bandwidth from every band $(\hat{R}_{\mathcal{B}} - \hat{R}_{\text{NULL}} |\mathcal{B}| / |\text{NULL}|)$, because a white cross-PCU correlated component was found (??).

1.10 NICER/XTI features (v9–v13)

1.10.1 Reading the events

Cleaned event files are read without HEASoft by a streaming FITS parser (`v9lib.PrefixParser`) that decompresses the gzip stream and extracts TIME, PI and DET_ID from the binary table rows. Three reading modes were used: v9 downloaded only the *prefix* of each file (HTTP range requests) until ≥ 6 gap-free 128 s segments, end of file, or 300 MB; v10 appended the *tail* up to 1 GB compressed and parsed prefix + tail as one stream; v11 and v13 *streamed* whole files (up to 1 GB) into memory without storing them. Parsing was checked to be identical to `astropy` and across the three modes (IC1 of v9–v11).

1.10.2 Good intervals, segments and detectors

Good intervals are maximal stretches without a gap $> 1 \text{ s}$ between consecutive events of any energy (`v9lib.good_intervals`). Each good interval $[a, b]$ is cut sequentially into 128 s segments

starting at a . Only 2–10 keV events are used for timing (PI 200–999, PI in units of 10 eV). The detector unit is the MPU, $m = \lfloor \text{DET_ID}/10 \rfloor \in \{0, \dots, 6\}$. Events are binned at $\Delta t = 1/256$ s ($N = 32768$, Nyquist 128 Hz). MPU m is on in a segment if each of the eight 16 s blocks has a 2–10 keV event; ≥ 2 MPUs are required. Segments overlapping a burst interval (Section 1.7) are removed.

1.10.3 Per-segment statistics and observation features

For each valid segment r (`v9lib.segment_stats`): s_m is the mean 2–10 keV counts per bin of MPU m (no background subtraction), the all-pairs cross-power is formed as in Eq. (1.17), and for each band $\text{num}_{\mathcal{B},r} = \frac{2}{N^2} \sum_{j \in \mathcal{B}} \text{cross}_j$, $\text{den}_r = \sum_{m < m'} s_m s_{m'}$. Observation features use the ratio of means (Eq. (1.20)) for T_1, T_2, T_3 , the state band 0.1–10 Hz and the 96–128 Hz white monitor, and the signed square root. In addition:

- **Colours** (`c1`, `c2`): counts of *all* events in the accepted segment windows in PI 200–399 (A , 2–4 keV), 400–599 (B , 4–6 keV) and 600–999 (C , 6–10 keV); $c_1 = B/A$, $c_2 = C/B$.
- **Count rate** `rate_2_10`: mean over segments of the 2–10 keV counts of the MPUs that are on, divided by 128 s.
- ν_c : 13 octave bins $k = 0, \dots, 12$ with $j \in [2^k, 2^{k+1} - 1]$ (1/128–64 Hz, Table 1.7); $\ell_k = \sum_r \text{num}_{k,r} / \sum_r \text{den}_r$; $\nu_c = \exp(\sum_k w_k \ln \nu_k / \sum_k w_k)$ with $w_k = \max(\ell_k, 0)$ and $\nu_k = 2^{k+0.5}/128$ Hz; missing if $\sqrt{\max(\sum_k \ell_k, 0)} < 0.05$.
- **State**: Eq. (1.19) applied to the 0.1–10 Hz rms.
- **Quality**: at least 3 valid segments and `rate_2_10` $\geq 20 \text{ cs}^{-1}$ (background then $\lesssim 5\%$ by the pre-registered estimate).

1.10.4 Background proxy (v10)

$\beta_i = N(\text{PI } 1200\text{--}1499)/N(\text{PI } 200\text{--}999)$, the 12–15 keV to 2–10 keV event ratio of the whole file (`v10lib.bkg_ratio`); the source contributes little above 12 keV, so a high β indicates background. The threshold is the 99th percentile of β among observations with `rate_2_10` $\geq 500 \text{ cs}^{-1}$ (where background is negligible); it was frozen at 0.2687 and reused in v11.

1.10.5 3C50 background correction and screening (v12, v13)

For observations with `rate_2_10` $< 500 \text{ cs}^{-1}$, HEASoft `nibackgen3C50` (?) was run inside WSL with default parameters and the NICER CALDB (`scripts/v12_bkg3c50.sh`); it returns a total spectrum $T(\text{PI})$ and a background spectrum $B(\text{PI})$. Band background fractions are $f_{\mathcal{X}} = \sum_{\mathcal{X}} B / \sum_{\mathcal{X}} T$ for $\mathcal{X} \in \{A, B, C, T = 200\text{--}999\}$. Brighter observations get $f = 0$. The frozen correction (`corrected` in `59_v12_analysis.py`, copied verbatim into `63_v13_analysis.py`) is

$$T'_k = \frac{T_k}{1 - f_T} \quad (k = 1, 2, 3, \text{state}), \quad c'_1 = \frac{B(1 - f_B)}{A(1 - f_A)}, \quad c'_2 = \frac{C(1 - f_C)}{B(1 - f_B)}, \quad (1.21)$$

followed by re-classification of the state; ν_c is not changed (it depends only on the shape of the power spectrum). The first relation holds because a fractional rms is normalised to the total mean; normalising to the source mean $S = (1 - f)(S + B)$ multiplies it by $1/(1 - f)$ (assuming a constant background). Screening keeps an observation if it is bright, or if 3C50 succeeded and $f_T \leq 0.20$. 3C50 failures (mostly data processed with the oldest gain calibration, which the tool rejects) are excluded. In v13 a descriptive variant recomputed the faint observations'

features from events inside the GTIs of the 3C50 total spectrum only (`gti_restricted` in `63_v13_analysis.py`).

1.10.6 Lorentzian fits (v11c)

The 13 octave-bin fractional variances ℓ_k (with delta-method errors, `v11lib.lb_se`) are fitted by one or two zero-centred Lorentzians whose integrated variance in a bin $[\nu_k^{\text{lo}}, \nu_k^{\text{hi}}]$ (bin edges $(j \mp 0.5)/128$ Hz) is

$$L_k(r^2, \Delta) = r^2 \frac{2}{\pi} \left[\arctan \frac{\nu_k^{\text{hi}}}{\Delta} - \arctan \frac{\nu_k^{\text{lo}}}{\Delta} \right], \quad \Delta \in [1/128, 128] \text{ Hz}, \quad (1.22)$$

by weighted least squares in log-parameters (`scipy.optimize.least_squares`, four starting points each, `v11lib.fit_lorentz`); $\text{BIC} = \chi^2 + k \ln n$ with $k = 2$ or 4 parameters; two components are preferred if $\text{BIC}_1 - \text{BIC}_2 > 6$. The width of the lowest-frequency component, ν_L , is the “break” frequency.

1.11 MAXI/GSC features (v7c, v8d, v9c)

The MAXI 1-day standard light curve of a source gives photon fluxes L (2–4 keV), M (4–10 keV), H (10–20 keV) and errors. Points are kept if every band has flux/error $\geq \sqrt{3}$ and positive errors, then points with $L + M + H \geq (\bar{L} + \bar{M} + \bar{H}) + 10 \text{ sd}$ are dropped; sources need ≥ 100 points (cuts of ?). Features per point (`39_v7c2_analysis.py`):

$$\text{SC} = \frac{M - L}{M + L}, \quad \text{HC} = \frac{H - L}{H + L}, \quad C_4 = \frac{H - M}{H + M}, \quad \text{RelInt} = \frac{L + M + H}{Q_{99.99}(L + M + H)}, \quad (1.23)$$

where $Q_{99.99}$ is the 99.99th percentile over that source’s points. Feature sets: $\text{CCI} = (\text{SC}, \text{HC}, \text{RelInt})$, $\text{C2D} = (\text{SC}, \text{HC})$, $\text{C1D} = (C_4)$ (the only colour without 2–4 keV).

N_{H} correction (v8d; `45_v8d_nh.py`, `v8d_heasoft.sh`). The Galactic HI column N_{H} at each source position is the HI4PI (?) weighted average within 0.1° (HEASoft `nh`, output `avwnh`). XSPEC computes photon fluxes of `tbabs*powerlaw` ($\Gamma = 2$; `abund wilm`, `xsect vern`; ?) on a grid $N_{\text{H}} \in \{0\} \cup \{10^{19}, \dots, 10^{23.5}\} \text{ cm}^{-2}$ (200 log steps); the transmission is $t_{\mathcal{X}}(N_{\text{H}}) = \Phi_{\mathcal{X}}(N_{\text{H}})/\Phi_{\mathcal{X}}(0)$ (e.g. 0.864, 0.979, 0.997 in L, M, H at 10^{22} cm^{-2}), interpolated linearly in $\log N_{\text{H}}$. Corrected fluxes $L' = L/t_L$, $M' = M/t_M$, $H' = H/t_H$ give SC' , HC' , C_4' , RelInt' by Eq. (1.23) (sets C2D_NH , C1D_NH , CCI_NH).

1.12 Feature summary

Table 1.8: Summary of every feature that enters a model or test. Missing-value rule: “excluded” = the observation does not enter; “median+ind.” = training-fold median and 0/1 indicator (??); “term = 0” = CCTLR timing term set to zero.

Feature	Unit	Definition	Code	Non-positive / missing
x_{ij} (rate)	$\text{cs}^{-1} \text{ keV}^{-1} \text{ Hz}$	$\text{RQU}(1.6)$	<code>04_fetch_process.py</code>	kept as is

Feature	Unit	Definition	Code	Non-positive / missing
A_{ij}	–	Eq. (1.9)	v2_representations	asinh defined for < 0
B_{ij}	keV $^{-1}$	Eq. (1.10)	same	$F > 0$ guaranteed
h_1, h_2	–	Eq. (1.11)	same	denominators > 0 ; numerators kept
$\log_{10} F$	dex	Eq. (1.7)	same	$F > 0$
B3, H3	–	Section 1.5	12_v3b_extended_band.py	as above
$R_{ij}, h_{1,2}^R$	–	Eq. (1.12)	reps_R	as above
X_1, X_2	–	Eq. (1.13)	19_v4b3_hexte.py	S/N $< 3 \rightarrow$ median+ind. (per colour)
T_1, T_2, T_3 (v5)	frac. rms	Eqs. (1.15) and (1.16)	v5lib.obs_timing	< 3 rows $\rightarrow$ median+ind.
T_k^* (v6)	frac. rms	Eq. (1.17)	v6lib.obs_timing_v6	median+ind.; CCTLR: term = 0
ν_c (RXTE)	Hz	Eq. (1.18)	v6lib.nu_centroid	missing $\rightarrow$ observation not in test
state rms (v7a)	frac. rms	Eq. (1.19)	v7lib.state_rms	< 3 rows $\rightarrow$ “unknown”
LC, HF1–3	frac. rms	Eq. (1.20)	v8lib.hf_features	QC fail $\rightarrow$ excluded
c_1, c_2 (NICER)	–	Section 1.10	v9lib.obs_features	excluded if QC fails
T_k , state (NICER)	frac. rms	Eq. (1.20)	same	excluded if QC fails
ν_c (NICER)	Hz	Section 1.10	same	missing $\rightarrow$ not in test
β	–	Section 1.10	v10lib.bkg_ratio	–
f_A, f_B, f_C, f_T	–	Eq. (1.21)	58_v12_bkg.py	3C50 failure $\rightarrow$ excluded
ν_L	Hz	Eq. (1.22)	v11lib.fit_lorentz	fit failure $\rightarrow$ missing
SC, HC, C_4 , RelInt	–	Eq. (1.23)	39_v7c2_analysis.py	cuts in Section 1.11
SC', HC', C'_4	–	Section 1.11	45_v8d_nh.py	–

Chapter 2

Phase 1

Bibliography